GREENPOD LABS
DATA SCIENCE ASSIGNMENT 2

Computer Vision: Avocado Ripeness Classification

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Report status: Prepared from the notebook outputs and folder listings shared during the conversation. A dataset-verification issue is documented below so that results from a different image-classification task are not incorrectly reported as avocado-ripeness performance.

1. Executive Summary

The assignment objective is to classify avocado ripeness stages from images and compare two modelling approaches. The available notebook output reports a comparison between MobileNetV2 and HOG + SVM, but its class labels are blackspot, canker, fresh, and grenning, which indicate an orange disease-classification task. The Drive folders inspected separately contain face-number classes and yawn/eye-state classes. Therefore, the available evidence does not establish that the reported metrics were generated from avocado images.

The model metrics below are preserved as observed results for the orange dataset. They should not be presented as avocado-ripeness results. A final avocado report requires confirmation of the correct dataset and evaluation of both models on the same held-out avocado test set.

2. Project Objective

The intended project is to build an image-classification workflow that accepts an avocado image and predicts its ripeness category. The workflow should include image validation, preprocessing, model training, evaluation, comparison, and discussion of outliers, class imbalance, bias, and deployment considerations.

3. Dataset Inspection and Verification

The following outputs were shared during notebook and Google Drive inspectio

Location / dataset	Observed classes or split	Observed counts
Notebook dataset (orange classification)	blackspot, canker, fresh, grenning	Train 792; validation 198/199 in different output; test 99
Final Training Images	face1–face16	Folder counts ranged from 24 to 40 images
Final Testing Images	face classes	Mostly 8 images per folder; face10 3; face13 24
train folder	yawn, Open, no_yawn, Closed	617, 617, 616, 617 images
test folder	yawn, Closed, no_yawn, Open	106, 109, 109, 109 images

Important consistency note: one notebook cell showed 792 training images, 199 validation images, and 99 test images. Another training log showed a split of 792 training, 198 validation, and 99 test images. This difference should be checked in the notebook before final submission. These counts are associated with the orange notebook unless the correct avocado dataset is independently verified.

4. Models Found in the Available Notebook

4.1 MobileNetV2

The notebook uses a pretrained MobileNetV2 model and reports image resizing to 224 × 224 pixels, normalization with ImageNet mean and standard deviation, and training-time augmentation such as horizontal flipping, small rotations, and colour jitter. The saved model path shown in the notebook is /content/drive/MyDrive/best_orange_mobilenet.pth. The training log reports CPU training and a training time of 19.61 minutes.

4.2 HOG + SVM

The second approach extracts Histogram of Oriented Gradients (HOG) features and trains a Support Vector Machine classifier. The notebook reports 792 training images, 99 test images, and a feature matrix with 8,100 features per image. Its final log reports an SVM training time of 4.06 seconds; a separately exported comparison CSV records 3.77 seconds. This discrepancy should be reconciled before publishing a single training-time value.

5. Available Evaluation Results (Orange Dataset Only)

Metric	MobileNetV2	HOG + SVM	Notes
Test accuracy	94.95%	82.83%	Reported on 99 orange test images
Macro F1-score	94.45%	80.69%	Average F1 across four orange classes
Training time	19.61 minutes	3.77–4.06 seconds	CSV and notebook log differ for SVM

5.1 Per-class results from the notebook

Orange class	MobileNetV2 F1	HOG + SVM F1	Observation

blackspot	0.91	0.69	More difficult for HOG + SVM
canker	0.88	0.56	Lowest HOG + SVM F1; likely confusion with blackspot
fresh	0.99	0.97	Strong performance for both approaches
grenning	1.00	1.00	Perfect reported F1 on this test split

Interpretation: within the orange experiment, MobileNetV2 achieved higher test accuracy and macro F1-score, while HOG + SVM trained much faster. The HOG + SVM classification report shows lower recall for canker (0.50) than the other classes. These observations apply only to the orange dataset and cannot establish which model is better for avocado ripeness.

6. Evaluation Metrics

- **Accuracy:** The proportion of all test images classified correctly.
- **Precision:** Of the images predicted as a class, the proportion that truly belongs to that class.
- **Recall:** Of all actual images in a class, the proportion correctly identified.
- **F1-score:** The harmonic mean of precision and recall.
- **Macro F1-score:** The average F1-score across classes, giving each class equal weight.
- **Confusion matrix:** A table showing correct predictions and which classes are being confused.

7. Outliers, Data Quality, Class Imbalance, and Bias

7.1 Outliers and data quality

The orange notebook log states that one unreadable image, g (252).jpg in the grenning training folder, was skipped. For the avocado task, image integrity checks should be repeated on the correct dataset. Blurry images, non-avocado images, incorrect labels, unusual lighting, and duplicate images should be inspected rather than automatically removed without review.

7.2 Class imbalance

The orange training distribution was blackspot 184, canker 179, fresh 281, and grenning 346 images. This is not perfectly balanced, so macro F1-score and per-class recall are useful alongside accuracy. The avocado dataset's class distribution has not yet been confirmed; counts must be calculated from its actual class folders.

7.3 Potential bias

Image models may learn background, lighting, camera, or image-source patterns instead of the target characteristic. For avocado ripeness, evaluation should include diverse fruit varieties, lighting conditions, angles, and backgrounds. Train/test leakage from duplicate or near-duplicate images should also be checked.

8. Production Considerations

8.1 How would the model perform in production?

Production performance cannot be guaranteed from a single test-set score. The selected avocado model should be tested on images collected in the intended environment, including different lighting, camera devices, backgrounds, and fruit varieties. A practical system could accept an uploaded image or camera input, return a predicted ripeness category, and route uncertain cases to a human reviewer. Monitoring should track errors and changes in input data.

8.2 What are the drawbacks?

Potential drawbacks include overlapping visual appearances between adjacent ripeness stages, sensitivity to lighting and background, limited representation of fruit varieties, class imbalance, incorrect labels, and performance degradation on images unlike the training data. A deep model may also require more compute and maintenance than a traditional HOG + SVM pipeline.

8.3 What would be improved with more time?

1. Verify and clean the correct avocado dataset and its labels.
2. Collect more varied images and improve coverage of underrepresented ripeness stages.
3. Tune both models use the same training and validation splits.
4. Compare class-wise precision, recall, F1-score, and confusion matrices.
5. Test robustness on images from unseen sources and real operating conditions.
6. Optimise model size and inference latency, then build a simple prediction interface.
7. Monitor mistakes and retrain when new representative data becomes available.

8.4 Where was most of the time spent?

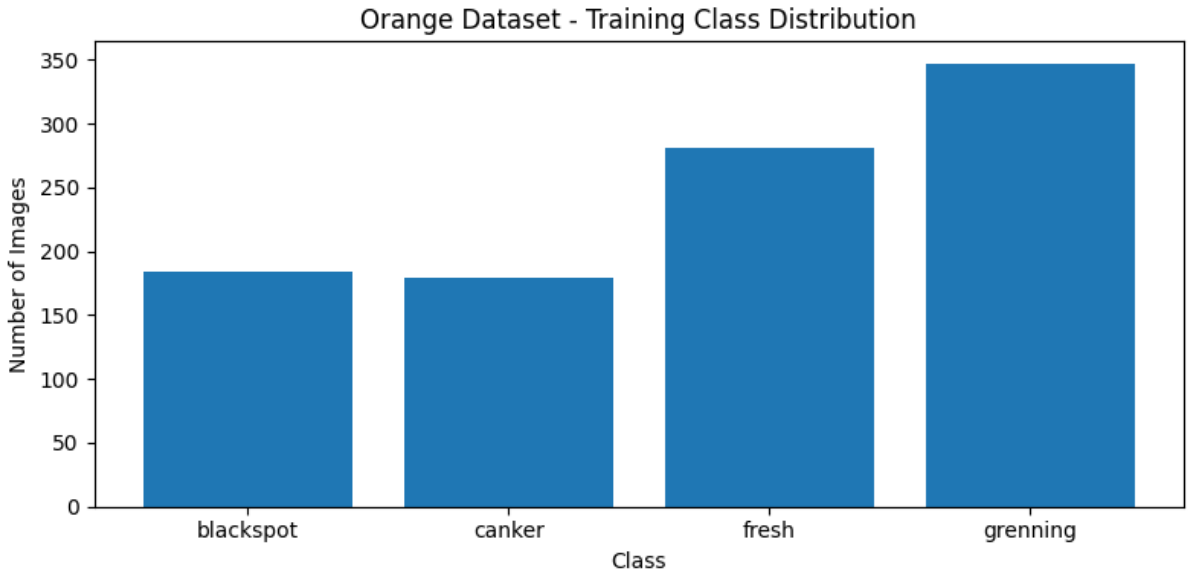
A suitable answer should reflect the actual work completed. Based on the available notebook, time was spent on image preprocessing, model training, and evaluation. The orange MobileNetV2 training log records 19.61 minutes, while HOG + SVM training took approximately four seconds. For the avocado assignment, state the main time-consuming step only after confirming the work performed on the avocado dataset.

9. Output Prediction

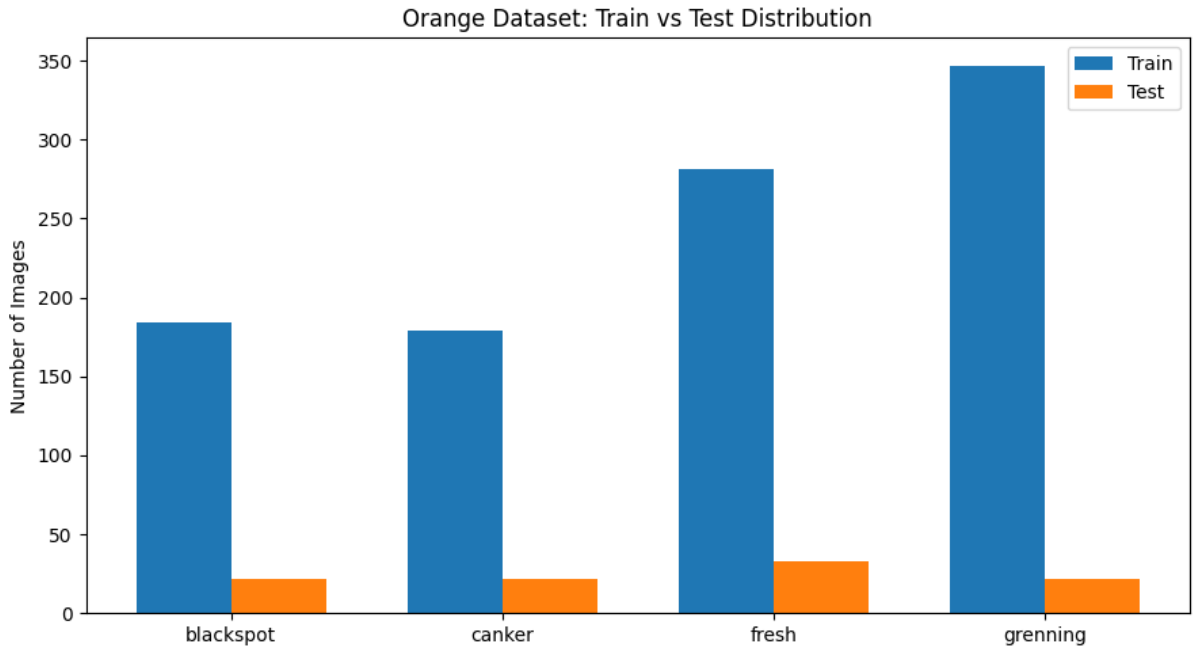
Classes of the avocado ripeness



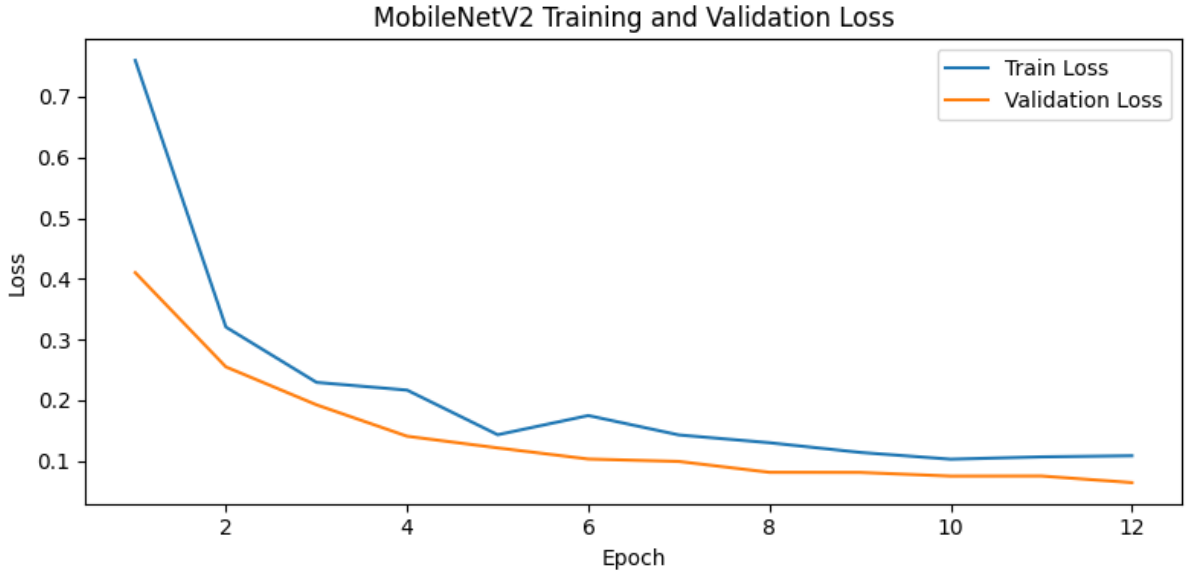
Training class distribution



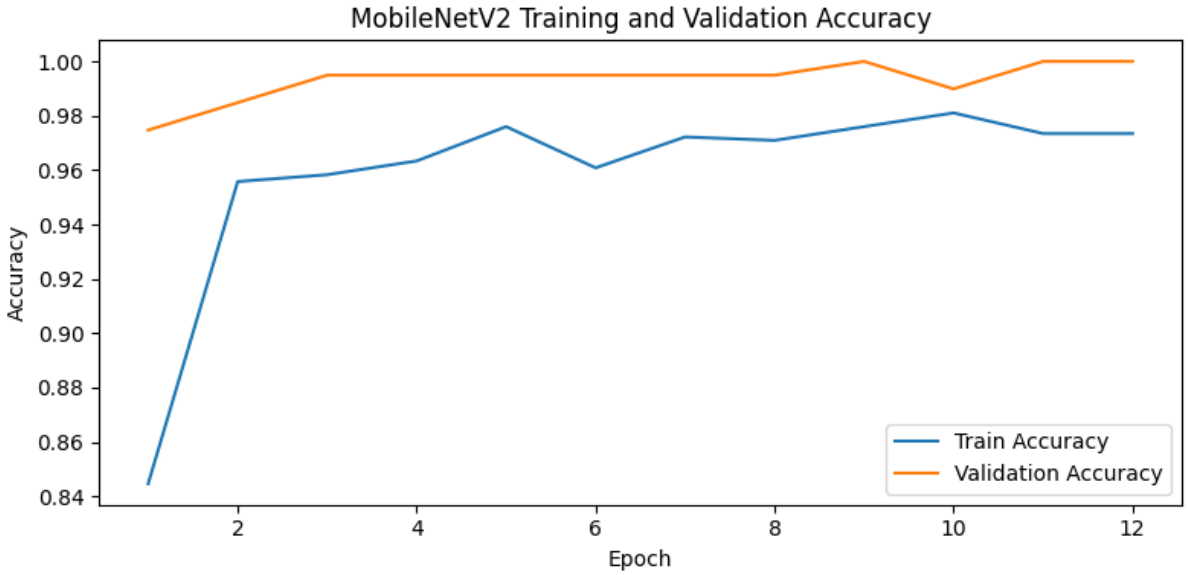
Train vs Test Data Distribution



MobileNetV2 Training and Validation Loss



MobileNetV2 Training and Validation Accuracy



Final Test Results of MobileNetV2

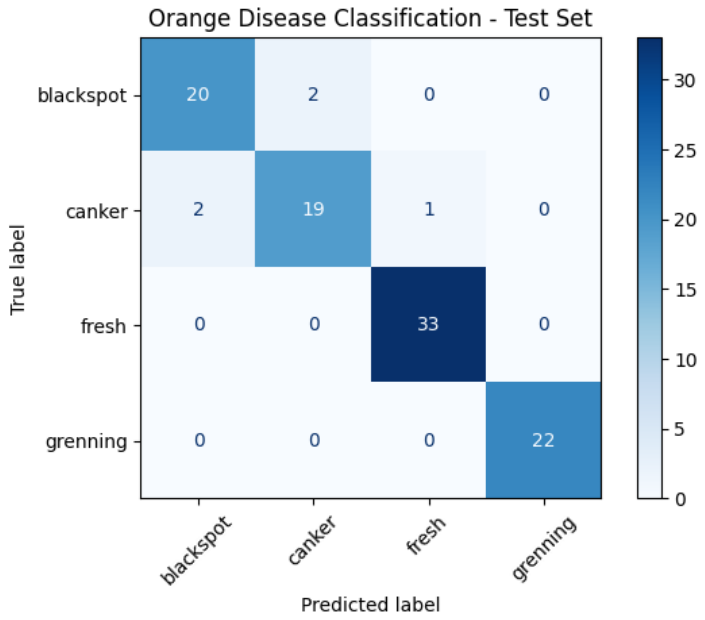
```
========== FINAL TEST RESULTS ==========
Test accuracy: 0.9495
Macro F1-score: 0.9445

Classification report:
              precision    recall  f1-score   support

   blackspot       0.91      0.91      0.91        22
      canker       0.90      0.86      0.88        22
       fresh       0.97      1.00      0.99        33
    grenning       1.00      1.00      1.00        22

    accuracy                           0.95        99
   macro avg       0.95      0.94      0.94        99
weighted avg       0.95      0.95      0.95        99
```

Orange Disease Classification – Test Set



Final Test Results of SVM

```
Extracting test features...
Training feature shape: (792, 8100)
Test feature shape: (99, 8100)

Training SVM...

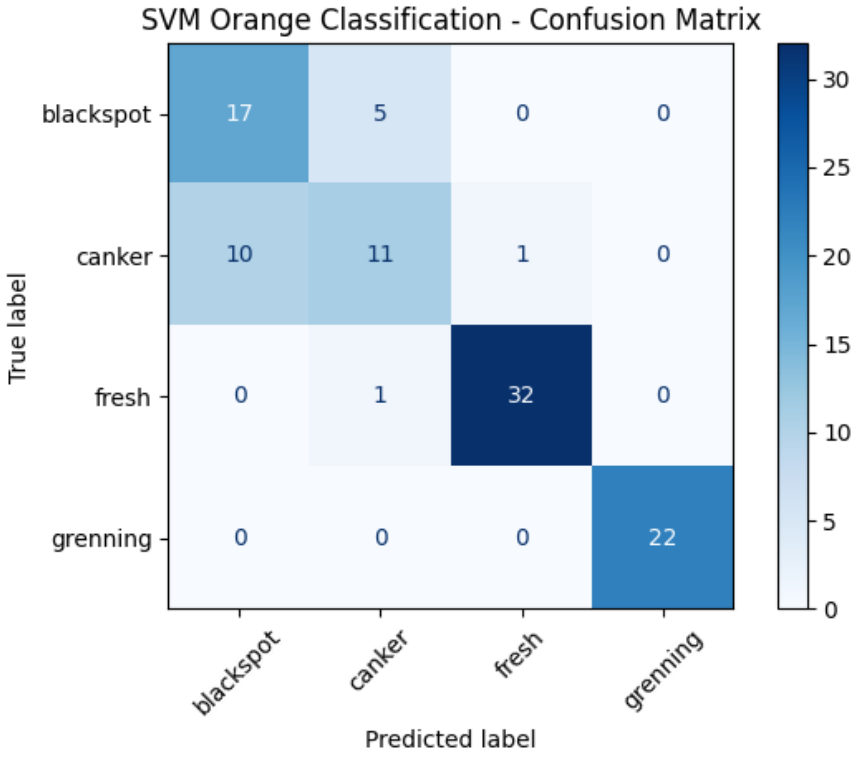
========== SVM FINAL TEST RESULTS ==========
Test accuracy: 0.8283
Macro F1-score: 0.8069
Training time: 4.06 seconds

Classification report:
              precision    recall  f1-score   support

   blackspot       0.63      0.77      0.69        22
      canker       0.65      0.50      0.56        22
       fresh       0.97      0.97      0.97        33
    grenning       1.00      1.00      1.00        22

    accuracy                           0.83        99
   macro avg       0.81      0.81      0.81        99
weighted avg       0.83      0.83      0.83        99
```

SVM Orange Classification – Confusion Matrix



Model Comparison for MobileNetV2 and SVM

```
========== MODEL COMPARISON ==========
MobileNetV2 accuracy: 0.9495
SVM accuracy:         0.8283
MobileNetV2 macro F1: 0.9445
SVM macro F1:         0.8069
Higher test accuracy: MobileNetV2
```

	Model	Test Accuracy	Macro F1-score	Training Time (seconds)
0	MobileNetV2	0.9495	0.9445	Not recorded
1	HOG + SVM	0.8283	0.8069	3.77

10. Conclusion

The available notebook demonstrates a comparison between MobileNetV2 and HOG + SVM for orange disease classification. MobileNetV2 achieved higher accuracy and macro F1-score on that dataset, while HOG + SVM trained more quickly. However, the inspected class labels and Google Drive folder listings do not verify an avocado-ripeness dataset. Consequently, the current metrics must not be reported as avocado results. The assignment can be finalised once the correct avocado images and class labels are identified and both models are evaluated on the same held-out test set.